



Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

LESSON PLAN

For

Wireshark Network Analysis Masterclass

Course Code: C1123

Conducted by

Tertiary Infotech Academy Pte Ltd

UEN: 201200696W

Version 5.1

DOCUMENT VERSION CONTROL RECORD

Version Number	Effective Date of Release	Summary of Included Changes	Author
4.0	2 October 2026	Revised from the supplied v3 deck; current guidance, visuals and 18 self-contained labs.	Dr. Alfred Ang
5.0	5 October 2026	Deck rebuilt on the house non-WSQ engine (C735 design system): full admin opener, core-concepts section, reference diagrams re-presented as captioned visuals, TShark evidence visuals per lab, per-topic schedule with day dividers and breaks.	Dr. Alfred Ang
5.1	5 October 2026	Lab step slides replaced by scenario + summary slides (full steps in the LG and new per-lab instruction MD/PDF). About 40 new concept slides with visuals drawn from the lab captures; DHCP, NAT, IPv6, TShark, security patterns and baselining added from external sources. Per-lab scenario tickets, findings templates, TShark equivalents and optional extensions.	Dr. Alfred Ang

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Course Information

Course Title	Wireshark Network Analysis Masterclass
Course Reference	C1123
Training Provider	Tertiary Infotech Academy Pte Ltd (UEN 201200696W)
Duration	4 days · 7.5 instructional hours per day (30 hours)
Daily Timing	9:30 am – 6:30 pm (1-hour lunch; tea breaks within training time)
Mode	Instructor-led demonstrations and practical labs; 30 instructional hours
Trainer	Dr. Alfred Ang

Learning Outcomes

On completion of this course, learners will be able to:

- LO1: Plan scoped captures and configure a reproducible analyst profile.
- LO2: Navigate, filter, summarize and time network exchanges.
- LO3: Interpret DNS, ARP, IPv4, ICMP and UDP evidence.
- LO4: Diagnose TCP and application behaviour, including HTTP and authorised TLS inspection.
- LO5: Report observed facts, hypotheses and next actions with defensible evidence.

Learning Reinforcement

- Every topic is followed by hands-on labs that learners complete on their own machine.
- Each lab ends with a 'Test it' verification step so learners can confirm their own progress.
- The trainer circulates during lab time to give individual feedback and unblock learners.
- Each topic closes with a recap mapped back to the course learning outcomes.

Course Schedule

Day 1 — Capture and analyst workflow

Time	Duration	Topic / Activity
9:30–9:50	20 min	Welcome, trainer and learner introductions, ground rules, course objectives and lab setup check
9:50–10:20	30 min	Topic 1 — Introduction to Network Analysis and Wireshark (concepts + demonstration)
10:20–11:00	40 min	Hands-on: Lab 1: Establish a trace baseline
11:00–11:15	15 min	Tea break
11:15–11:45	30 min	Topic 2 — Capture Methods and Capture Filters

		(concepts + demonstration)
11:45–13:00	75 min	Hands-on: Lab 2: Plan capture placement
13:00–14:00	60 min	Lunch break
14:00–14:30	30 min	Topic 3 — Global Preferences and Troubleshooting Profiles (concepts + demonstration)
14:30–15:30	60 min	Hands-on: Lab 3: Create an analyst profile
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Topic 4 — Navigation and Coloring Techniques (concepts + demonstration)
16:15–17:30	75 min	Hands-on: Lab 4: Color and annotate evidence
17:30–18:00	30 min	Guided practice: re-run Labs 1–4 with your own analyst profile; worked examples
18:00–18:30	30 min	Day 1 recap, learning reinforcement and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 2 — Time, statistics, filters and DNS

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 1 review and Day 2 objectives
9:45–10:10	25 min	Topic 5 — Time Values and Delay Types (concepts + demonstration)
10:10–11:00	50 min	Hands-on: Lab 5: Separate path and server delay
11:00–11:15	15 min	Tea break
11:15–11:40	25 min	Topic 6 — Trace Statistics and VoIP Overview (concepts + demonstration)
11:40–12:30	50 min	Hands-on: Lab 6: Summarize traffic and a voice stream
12:30–13:00	30 min	Guided practice: timing and statistics worked examples (Labs 5–6)
13:00–14:00	60 min	Lunch break
14:00–14:15	15 min	Topic 7 — Display Filters (concepts + demonstration)
14:15–14:45	30 min	Hands-on: Lab 7: Build a filter evidence matrix
14:45–15:00	15 min	Topic 8 — TCP/IP Communications and Resolution (concepts + demonstration)
15:00–15:30	30 min	Hands-on: Lab 8: Reconstruct an application dependency chain
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Topic 9 — DNS Traffic Analysis (concepts +

		demonstration)
16:15–17:15	60 min	Hands-on: Lab 9: Diagnose DNS failure and delay
17:15–18:00	45 min	Guided practice: filter, resolution and DNS worked examples (Labs 7–9)
18:00–18:30	30 min	Day 2 recap, learning reinforcement and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 3 — IP and transport diagnostics

Time	Duration	Topic / Activity
9:30–9:45	15 min	Day 2 review and Day 3 objectives
9:45–10:10	25 min	Topic 10 — ARP Traffic Analysis (concepts + demonstration)
10:10–11:00	50 min	Hands-on: Lab 10: Verify link-local resolution
11:00–11:15	15 min	Tea break
11:15–11:40	25 min	Topic 11 — IPv4 Traffic Analysis (concepts + demonstration)
11:40–12:30	50 min	Hands-on: Lab 11: Classify IPv4 scope and headers
12:30–13:00	30 min	Guided practice: ARP and IPv4 worked examples (Labs 10–11)
13:00–14:00	60 min	Lunch break
14:00–14:15	15 min	Topic 12 — ICMP Traffic Analysis (concepts + demonstration)
14:15–14:45	30 min	Hands-on: Lab 12: Interpret echo and unreachable messages
14:45–15:00	15 min	Topic 13 — UDP Traffic Analysis (concepts + demonstration)
15:00–15:30	30 min	Hands-on: Lab 13: Follow datagrams and service refusal
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Topic 14 — TCP Protocol Analysis (concepts + demonstration)
16:15–17:15	60 min	Hands-on: Lab 14: Investigate retransmission and zero window
17:15–18:00	45 min	Guided practice: transport-layer worked examples (Labs 12–14)
18:00–18:30	30 min	Day 3 recap, learning reinforcement and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Day 4 — Graphs, applications and reporting

Time	Duration	Topic / Activity
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9:30–9:45	15 min	Day 3 review and Day 4 objectives
9:45–10:10	25 min	Topic 15 — Traffic Graphs (concepts + demonstration)
10:10–11:00	50 min	Hands-on: Lab 15: Graph bytes, RTT and recovery
11:00–11:15	15 min	Tea break
11:15–11:45	30 min	Topic 16 — HTTP and HTTP/2 Analysis (concepts + demonstration)
11:45–13:00	75 min	Hands-on: Lab 16: Inspect HTTP outcomes and HTTP/2 frames
13:00–14:00	60 min	Lunch break
14:00–14:30	30 min	Topic 17 — TLS-Encrypted Traffic Analysis (concepts + demonstration)
14:30–15:30	60 min	Hands-on: Lab 17: Compare encrypted and decrypted views
15:30–15:45	15 min	Tea break
15:45–16:15	30 min	Topic 18 — Ten Troubleshooting Steps and Reporting (concepts + demonstration)
16:15–17:30	75 min	Hands-on: Lab 18: Produce an evidence-led incident report
17:30–18:00	30 min	Synthesis: present your evidence-led incident report findings
18:00–18:30	30 min	Course recap, next steps and Q&A

Total scheduled time: 480 minutes (8 hours) excluding lunch — of which 450 minutes (7.5 hours) are instructional, plus 30 minutes of tea breaks.

Lab Reference (aligned to course topics)

Topic	Coverage	Labs
Topic 01: Introduction to Network Analysis and Wireshark	—	Lab 1
Topic 02: Capture Methods and Capture Filters	—	Lab 2
Topic 03: Global Preferences and Troubleshooting Profiles	—	Lab 3
Topic 04: Navigation and Coloring Techniques	—	Lab 4
Topic 05: Time Values and Delay Types	—	Lab 5
Topic 06: Trace Statistics and VoIP Overview	—	Lab 6
Topic 07: Display Filters	—	Lab 7
Topic 08: TCP/IP Communications and Resolution	—	Lab 8
Topic 09: DNS Traffic Analysis	—	Lab 9

Topic 10: ARP Traffic Analysis	—	Lab 10
Topic 11: IPv4 Traffic Analysis	—	Lab 11
Topic 12: ICMP Traffic Analysis	—	Lab 12
Topic 13: UDP Traffic Analysis	—	Lab 13
Topic 14: TCP Protocol Analysis	—	Lab 14
Topic 15: Traffic Graphs	—	Lab 15
Topic 16: HTTP and HTTP/2 Analysis	—	Lab 16
Topic 17: TLS-Encrypted Traffic Analysis	—	Lab 17
Topic 18: Ten Troubleshooting Steps and Reporting	—	Lab 18